

Networked Microgrids for Enhancing the Power System Resilience

This paper discusses how to form a networked microgrid cluster for improving power system resilience.

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ABSTRACT | This paper focuses on the role of networked microgrids as distributed systems for enhancing the power system resilience against extreme events. Resilience is an intrinsically complex property which requires deep understanding of microgrid operation in order to respond effectively in emergency conditions. The paper first introduces the definition and offers a generic framework for analyzing the power system resilience. The notion that large power systems can achieve a higher level of resilience through the deployment of networked microgrids is discussed in detail. In particular, the management of networked microgrids for riding through extreme events is analyzed. In addition, the merits of advanced information and communication technologies (ICTs) in microgrid-based distributed systems that can support the power system resilience are presented. The paper also points out the challenges for expanding the role of distributed systems and concludes that networked microgrids in particular provide a universal solution for improving the resilience against extreme events in Smart Cities.

KEYWORDS | Distributed power systems; hierarchical control of hybrid AC/DC microgrids; information and communication technologies; networked microgrids; resilience indices; situational awareness

I. INTRODUCTION

A more intense climate change has increased the frequency and the severity of weather-related events throughout the world. Within the first seven months in 2016, there were eight weather-related events (including two major flood and six severe storms) in the United States, each with financial losses exceeding \$1 billion [1]. In addition, since 2002, severe weather events have caused over 80% of major power outages involving 50 000 or more customers in the United States [2].

The alarming news is that the electric power infrastructure remains to be susceptible to disastrous events whose destructive power can spread in such a quick and wide manner that can initiate extensive damages and subsequent outages to large power systems [3]. For instance, Hurricane Sandy struck a large portion of the power system in the eastern United States at the end of October 2012 when nearly 7.5 million of electricity consumers lost power across 15 states. Even after a few days of power restoration initiatives, there were still more than 3.7 million consumers who remained without power [4]. These outages reveal the fact that power systems urgently need the additional capabilities and practical solutions to withstand such infrequent events which often have a very high impact on our societal undertakings [5].

Traditionally, power systems have been designed in accordance with reliability requirements, namely, the

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“N-1” or “N-2” criterion, which means the power systems are designed and operated to be able to absorb the failure of a single component or even the simultaneous failures of two single components. In other words, power system operations have already been considered robust for a selected number of low-impact/highly-probable contingencies. Although power systems are often designed to withstand such contingencies, neglecting the catastrophic (high-impact/low-probable) events that are less likely to happen renders them inadequately prepared for alleviating the whims of Mother Nature.

In practice, component failure analyses would be much more complicated considering severe weather events, as these components may trigger cascading failures. In such cases, multiple components are likely to fail in a quasi-simultaneous manner, which could leave electricity consumers to endure prolonged blackouts. Given the aging electric power infrastructure in the United States, the situation may even deteriorate more quickly in such occasions. Hurricane Sandy, for instance, was an “N-90” contingency [6]. Power systems that are inadequately prepared against various abnormal conditions, including both low-impact/highly-probable and high-impact/low-probable contingencies, are often doomed to lose the fight against severe weather events that stem out of climate changes.

It is a major challenge for power system operations to withstand unplanned catastrophic events [7]. Power outages in the wake of severe weather events often spur sufficient interests among utility planners in reinforcing the electric power infrastructure. Hardening power systems by physically altering the electricity infrastructure (e.g., utilizing underground cables, strengthening line towers, fortifying substations) is commonly deemed as one of the most effective methods to combat severe events [8]. The hardening measures can make power components less susceptible to severe outages, but power system enhancement cannot solely depend on infrastructure reinforcements. Even if all components across power systems are hardened, there is still no guarantee that they will stay intact during severe conditions [9], irrespective of the level of expenditure on hardening the infrastructure.

Although continuous efforts have been devoted to strengthening the electricity infrastructure, power system operators must often resort to additional actions to deal with unexpected power outages. In other words, as catastrophic events can hardly be prevented, human operators are required to plan for and respond to potential disruptions sufficiently so that the power systems can successfully ride through and recover from such events. As it is essential to keep the critical loads (e.g., hospitals, police stations and security buildings, data centers, shelters and rescue centers, major traffic lights) intact during emergencies, the deployment of distributed power systems such as microgrids is a viable option to retain the supply of power and ease the individual citizens’ discomfort caused by disruptive events [10]. Notably, the local power supply within a microgrid can be strategically managed and critical loads can be tactically identified and maintained, even when the remainder of the system is suffering from severe outages. For instance, when the electric utility services failed during Hurricane Sandy,

a few university campuses with cogeneration facilities in the stricken region remained energized by operating the campuses as islanded microgrids [11]. Such disasters and the corresponding consequences can have profound effects on understanding the role of microgrids in managing regional operations in emergency conditions [12].

The rest of the paper provides insights into the nature of the power system resilience (Section II), why power systems can rely on microgrid-based distributed systems for resilience enhancement (Sections III and IV), how microgrids ride through various extreme events to ensure resilience (Section V), how ICTs play a role in supporting resilience (Section VI), and what are the obstacles to realizing the full potential of microgrids for resilience enhancement (Section VII).

II. UNDERSTANDING POWER SYSTEM RESILIENCE

A. Definition of Power System Resilience

Considering that there is no universally-accepted definition of the power system resilience, we resort to the definition that takes into account the ability of a power system to prepare adequately for, respond comprehensively to, and recover rapidly from major disruptions due to extreme events. The inherent density of power distribution systems renders themselves vulnerable to intensive disruptions. About 90% of blackouts in the United States are initiated from power distribution system, which has proved to be the most vulnerable part of the electric power infrastructure [13]. In essence, the power system resilience is deemed to be equivalent to the resilience of power distribution system (abbreviated as resilience).

A resilient power system will be able to withstand disruptions pertaining to severe weather and climate changes (e.g., hurricanes, tornados), catastrophic man-made incidents (e.g., malicious attacks, human operator missteps), and a combination of such incidents. Table 1 lists several extreme events with the associated dates for blackouts.

Obviously, these rare events have disastrous effects, triggering a significant decline in the power system performance. The resulting power system disruptions present the following features that distinguish them from typical power system contingencies [2], [8], [22]:

Table 1 Representative of Extreme Events and the Resulting Blackouts

Extreme Event	Origin		Blackout	
	Natural	Man-made	Date	Location
Hurricane [14]	✓		Oct. 2012	U.S.
Wind Storm [15]	✓		Feb. 2008	China
Solar Flare [16]	✓		Mar. 1989	Canada
Flooding [17]	✓		Aug. 2005	U.S.
Earthquake [18]	✓		Mar. 2011	Japan
Thunderstorm [19]	✓		Aug. 2006	Canada
Cyberattack [20]		✓	Dec. 2015	Ukraine
Equipment Failure [21]		✓	Jul. 2012	India
Technical Error [22]		✓	Aug. 2003	U.S.

- the time of occurrence and duration of such events are highly uncertain which cannot be effectively predicted;
- the events often pose correlated spatiotemporal effects on the power system performance;
- they can disrupt any power system components and make repair and restoration processes very complex and time consuming;
- they can cause extensive component failures within a short time, and leave a large number of electricity consumers without power for an extended period;
- they often cause significant damages across the utility territory and have a major impact on other critical infrastructures (e.g., natural gas system, communication network, transportation system, water system).

B. Elements of Power System Resilience

Fig. 1 shows the composition of resilience against extreme events. In general, resilience comprises the following elements.

- Continuous situational awareness. Utilizing internal factors (e.g., power system operation and control strategies) and external conditions (e.g., dynamic weather situation), situational awareness enables the comprehension of the current situation, and even the projection of potential disruptions. In consequence, the extreme event can be managed proactively rather than in a passive way.
- Robustness and preparedness before any extreme event. Both infrastructural and operational measures are deployed in advance of extreme events so as to limit the impact of any potential disruptions. The electricity infrastructure is reinforced so that it will be resistant to postulated disruptions and less vulnerable to unexpected disruptions. In addition, power system operations ought to be flexible to absorb major disruptions. These resilience measures must adapt themselves to evolving extreme events.
- Responsiveness and survivability during any extreme event. Power system operators should respond to disruptions in a timely and effective manner so as to preserve the system functionality and curb any decline in the system performance. More importantly, power system must be able to survive any extreme events by maintaining a minimum level of functionality.
- Recoverability and rapidity after any extreme event. System performance must be recoverable and restored quickly to the level prior to the occurrence of extreme event.

C. Reliability Versus Resilience

The definition of resilience is distinguished from that of reliability. Conceptually, reliability refers to the system ability to consistently supply an adequate level of electricity

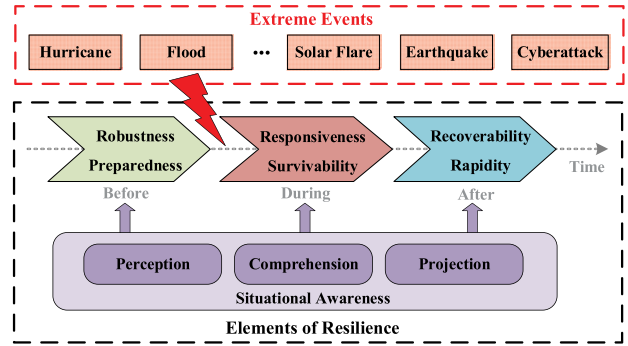


Fig. 1. Conceptual composition of power system resilience.

services to consumers [24]. The main differences between reliability and resilience are summarized as follows.

- Reliability measures the power system performance against potential low-impact/highly-probable events (e.g., $N - 1$ or $N - 2$ contingencies), whereas resilience corresponds to extreme events that have a low probability but a high impact (e.g., $N - k$ contingencies). Accordingly, resilience focuses more on the consequences of disruptions than their possibilities.
- In contrast to resilience, reliability is a merit-based feature, which can be more easily predicted by analyzing a case-by-case property of a power system. In other words, reliability can be evaluated without specifying the threats, but resilience is always relative to a particular threat. In fact, the more an engineering system is designed to be resilient to a specific type of extreme events, the more it will become fragile and vulnerable to some other types [24], [25]. For instance, burying overhead power lines under ground is helpful in achieving a higher level of resilience to hurricanes but, at the same time, makes the system more vulnerable to flooding.
- The reliability of a power system is commonly measured by the frequency and duration of power outages (e.g., lasting longer than 5 min [27]) experienced by electricity consumers. However, the evaluation of resilience focuses on changes in the system performance. In other words, resilience looks more deeply into how power systems are operated before, during, and after the occurrence of an extreme event.

There have been a number of standard indices for measuring the power system reliability. The commonly-used reliability indices include loss-of-load probability (LOLP), expected-energy-not-supplied (EENS), system-average interruption-frequency index (SAIFI), and the system-average interruption-duration index (SAIDI). However, there is a lack of resilience-based metrics which can be truly differentiated from the reliability indices [28]. In addition, a power system with a high reliability performance is not necessarily resilient. In a highly automated power system, faulty components can be removed within a sufficiently short time,

resulting in a reasonably high level of reliability. However, the same automated devices would have probably failed to work when experiencing an extreme event and therefore operating conditions would continue to deteriorate, in which case resilience is challenged. Hence, resilience and reliability are two complementary attributes desired to be implemented by modern power systems, both of which are indispensable for sustaining the power supply to electricity customers under any level of severity.

D. Resilience Indices

As power systems are continually threatened by extreme events under changing environmental and operational conditions, an urgent need exists to enhance the power system resilience to such events in order to avoid disastrous power outages. Initially, the resilience level should be properly quantified so as to guide enhancement efforts and evaluate their effectiveness in a rational manner. Resilience pertaining to a specific extreme event is essentially portrayed by changes in system performance after such an event unfolds. System performance itself can be converted into a quantifiable nonnegative value by considering a certain aspect of operations (e.g., generation adequacy [29], operation security [30], and power quality [31]). In principle, a higher level of system performance is more desirable.

Fig. 2 depicts a typical evolution curve of system performance in the wake of an extreme event, where the level of system performance $Q(t)$ is denoted as a function of time t . After the occurrence of an extreme event at time t_0 , the system performance remains at the normal level Q_0 due to the resistance to initial disruptions, and then it witnesses a rapid decline from time t_1 when disruptions become severer. At time t_2 , the system performance bottoms out at the lowest level $Q_{\min}$ and starts to bounce back starting from time t_3 which is due to ongoing restoration efforts. The system performance is below the normal level until time t_4 when the restoration is completed.

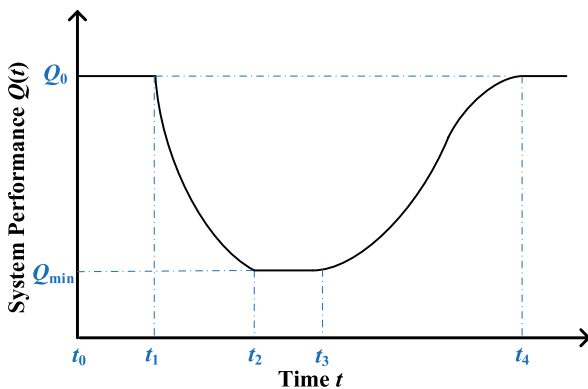


Fig. 2. Typical evolution curve of system performance.

The system resilience concerning this extreme event can then be quantified as the reciprocal of the system's loss of performance

$$\text{resilience} = \frac{1}{\text{loss}}.$$

The quantification of performance loss can be achieved by various means. Three options are given as follows.

- 1) Performance loss can simply be determined by its largest deviation from normal level, expressed as

$$\text{loss} = \frac{Q_0 - Q_{\min}}{Q_{\min}}$$

which directly reflects the system's survivability against the extreme event.

- 2) Performance loss can also be calculated by the integration of the relative deviation over the duration of performance degradation, i.e., from time t_1 when the extreme event occurs to time t_4 when the normal level performance (or an acceptable level of performance) is restored. That is

$$\text{loss} = \int_{t_1}^{t_4} \left[\frac{Q_0 - Q(t)}{Q(t)} \right] dt$$

which shows the extreme event's influences on the system performance from a more comprehensive perspective by capturing not only the degree of robustness and responsiveness to disruptions but also the swiftness of recovery [31], [32].

- 3) Furthermore, the duration of performance degradation can be considered explicitly to emphasize the importance of rapid recovery, expressed as

$$\text{loss} = \frac{1}{t_4 - t_1} \int_{t_1}^{t_4} \left[\frac{Q_0 - Q(t)}{Q(t)} \right] dt.$$

Hence, the degree of resilience can be quantified, which theoretically ranges from 0 to infinity. Specifically, infinity in this case refers to the perfect resilience (i.e., no performance degradation will be caused by the extreme event) and 0 means no resilience (i.e., failure to survive or even immediate collapse after the extreme event occurs).

E. Generic Framework for Resilience Analyses

On one hand, extreme events influence the system performance by damaging the electricity infrastructure and delaying its repairs for returning to normal operation. On the other hand, the system performance hinges on how human operators respond to such events. In order to dissect the resilience to an extreme event, a generic framework for resilience analyses is developed as shown in Fig. 3. This framework integrates the spatiotemporal characteristics of potential disruptions with infrastructural properties and emergency management effectiveness of power systems, regardless of the type of extreme events. The framework for a closed-loop performance considered in Fig. 3 also realizes the ongoing process of resilience assessment

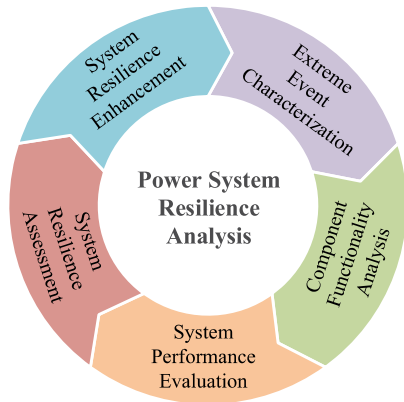


Fig. 3. Generic framework for analyzing the power system resilience.

and enhancement in order for power systems to adapt to dynamic conditions through learning from past experiences and facing evolving threats.

The proposed framework allows resilience analyses to follow the way extreme events affect power systems. First, all attributes of the extreme event (e.g., type, disruptive pattern, magnitude, duration) must be stated before analysts can embark on any kind of investigations on its implications. The individual power systems demonstrate a varying degree of vulnerability to different types of extreme events. So it is highly possible that a power system that is considered robust to natural hazards would be regarded as fragile to man-made errors. Furthermore, the spatiotemporal dynamics of the extreme event (e.g., intensity, geographical scope) ought to be adequately defined as the event moves across the system. The progression of a potential extreme event can normally be envisioned with reference to its historical counterparts. For instance, footprints of past severe weather events can be retrieved from the database of the National Oceanic and Atmospheric Administration (NOAA) [34]. Particularly, such inherent stochastic behaviors can be modeled as a scenario tree [35], where a tree-like collection of scenarios are formed from the occurrence of the extreme event and individual scenarios are analyzed separately.

Extreme events would damage power system components during their progression, either directly (e.g., striking down) or indirectly (e.g., triggering a cascading failure). The affected components may thus experience a complete (e.g., power line tower collapse caused by floodwater force) or partial failure (e.g., power line rating reduction as a result of wildfire heat). Practically, the influence of an extreme event on individual power system components can be translated into the reduction of their functionality (e.g., power line rating, generator capacity), which can be assessed by a time-dependent probability density function resulting from uncertainties embedded in the severity of disruptions.

In principle, each probability density function is jointly determined by multiple factors (e.g., component fragility, current intensity of the extreme event, cumulative effects

of the extreme event, support of maintenance crew). Every component's functionality is dynamically updated by probabilistic analyses after the occurrence of an extreme event, when the functionality's probability density function evolves with the progression of the extreme event. In practice, the components may suffer from continuous functionality degradation or even a total loss of functionality (i.e., complete failure) when exposed to intensive disruptions. If a component's functionality is severely distorted, the functionality will not rebound until enough efforts are exerted on the component restoration.

Concurrently, the level of infrastructure integrity is evaluated through the integration of relevant information on component functionalities. In this way, the impact of an extreme event on the system infrastructure is captured more accurately as compared with applying the static $N - k$ cardinality description along the extreme event's progression path [36], [37]. The system performance is then represented as a time-varying function of the infrastructure integrity level, which also incorporates the consequences of the operational actions in response to the unfolding disruptions. That is to say, the system performance during the extreme event is conditional on both the robustness of the system infrastructure and the effectiveness of operational counteractive measures.

Considering the temporal correlations of the system performance, a practical way to analyze resilience is by means of time-domain simulation (e.g., sequential Monte Carlo method [38], [39]). At each simulation step, the probability density functions representing components' functionality are obtained by utilizing an expert's knowledge or a data-driven approach, with the consideration of the progression of an extreme event along its path. The results of functionality analyses of individual components collectively determine the integrity level of the system infrastructure. Meanwhile, operators' responses are fully modeled including both the measures (e.g., emergency dispatch, power restoration) and their sequences. Accordingly, the system performance is evaluated by a sophisticated combination of infrastructure integrity and operators' responses. The system performance is continuously recorded until the restoration of its normal level. The evolution curve of the system performance indicates how the system behaves in the wake of an extreme event, and lays the foundation for resilience assessment. Eventually, resilience is assessed by measuring consequential shortcomings in the system performance. When multiple scenarios in an extreme event are considered, resilience assessment is first performed for each realization of the scenarios and the outcomes are merged following a predefined procedure.

The proposed framework shown in Fig. 4 helps determine if the system is sufficiently resilient for the designated extreme event. Fig. 4 also illustrates how enhancement measures affect the power system resilience under the proposed framework, where the solid and dotted lines represent

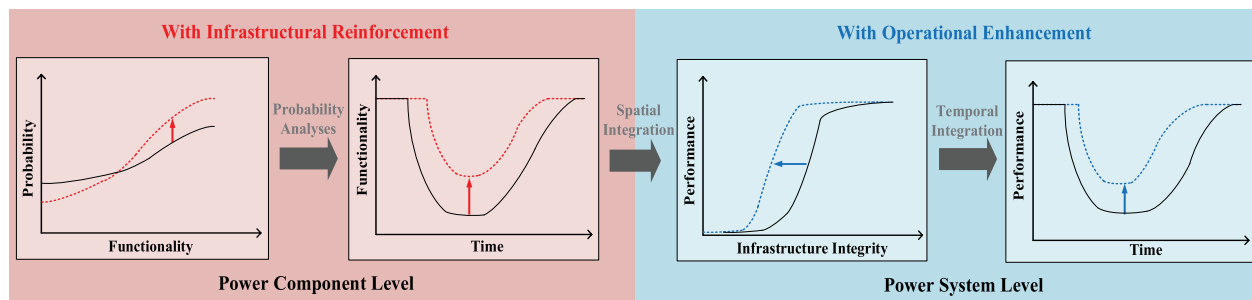


Fig. 4. Conceptual analyses of the power system resilience.

the associated functions before and after the provision of enhancement measures, respectively. After the estimation of potential consequences of an extreme event on the system performance, vulnerable system components and deficient operational procedures are identified and additional enhancement measures are effectively deployed to reinforce the system characteristics.

Conceptually, infrastructural reinforcement measures make power systems more resistant to disruptions by preserving individual component functionalities. On the other hand, operational enhancement measures provide a capability for degrading the system performance gracefully and returning it to the normal level rapidly. Comparative simulation can be conducted for pinpointing the bottlenecks in resilience improvement and recognizing the effectiveness of enhancement measures.

III. MIGRATING TOWARD DISTRIBUTED SYSTEMS

A. Features of a Microgrid

Distributed energy resources (DERs) dispersed in power distribution systems are gradually invading the territory once seized by centralized and large-scale generating facilities, mainly due to concerns with environmental pollution and the loss of traditional economies of scale [40]. DERs include microgenerators (e.g., diesel generator, fuel cell, photovoltaic panel, wind turbine) and energy storage devices (e.g., flywheel, supercapacitor, battery). However, the fast growth in the utilization of diverse DERs poses several challenges in the operation of power distribution systems. For one thing, the aging electricity infrastructure can hardly accommodate a high penetration level of DERs without a fundamental change in its hierarchy. For another, the real-time and centralized monitoring and control of a large number of DERs introduce several technical issues. In this context, a viable option to manage a power distribution system is to break it up into a set of smaller distributed systems, each with full observability and direct controllability over its domain. In this sense, microgrids represent a promising platform for developing distributed power systems to take full advantage of a multitude of DERs [41], [42].

A microgrid, characterized as the building block of Smart Grid, is a small-scale self-controllable power system interconnecting DERs and loads within clearly defined electrical boundaries [43], [44]. Microgrids interact with the main grid (e.g., service area of a utility) through points of common coupling (PCCs) at their boundaries. Microgrids can be operated in two modes, i.e., grid-connected mode when functioning in coordination with the external power system, and island mode when functioning independently, with the capability of a seamless transition between the two modes.

Microgrids help realize the benefits of local renewable energy sources through the utilization of the state-of-the-art power electronics-based control and demand-side management that can respond effectively to the mitigation of environmental pollution and climate changes [45]. Microgrids play a vital role in linking local consumers to the main grid [46]. Considering the main grid, each microgrid is treated as an aggregated and controllable load (i.e., either positive or negative) that provides more flexibility to the operation of variable power systems. From consumers' viewpoint, microgrids improve the power quality, lower electricity costs, and sustain power supply to critical loads in emergency conditions. The deployment of microgrids paves the way toward attaining resilient power systems. Microgrids can dramatically improve the survivability of the local power supply when the main grid suffers prolonged power outages.

A microgrid controller (MGC) is deployed for managing each microgrid strategically under various operating conditions [4], [47], [48]. Fig. 5 shows the typical functionality of an MGC. A number of local controllers (LCs) supervised by MGC are implemented in the microgrid for realizing finer grained monitoring, control, and management functions. Under the control of LCs, DERs can be further classified as grid-following and grid-forming DERs. Grid-following DERs are usually nondispatchable (e.g., photovoltaic panel, wind turbine) and can follow microgrid voltages and frequency rather than directly control them, whereas grid-forming elements are usually dispatchable (e.g., generators, energy storage devices) with a sufficient capacity for real and reactive power to regulate microgrid voltages and frequency.

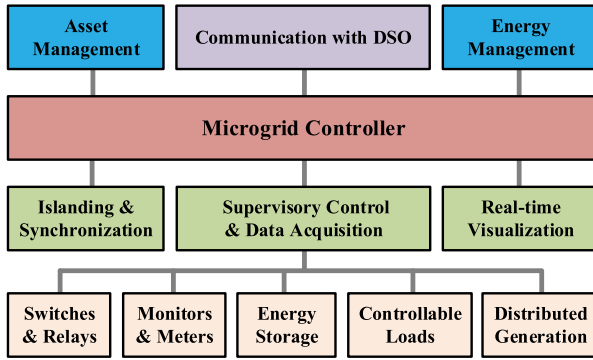


Fig. 5. Functionality of a microgrid controller.

B. Microgrid-Based Distributed Systems

As the generation supplied through each PCC is often limited to a few mega volt-ampere (MVA) [49], a power distribution system can host a number of microgrids that are networked to form a microgrid cluster. In practice, a microgrid can comprise several laterals of a distribution feeder or even a full distribution feeder. Fig. 6 gives an example of a set of networked microgrids which is based on the IEEE 13-bus test feeder. Microgrids within the same power distribution system could be centrally managed by a distribution system operator (DSO). The DSO supervises and coordinates MGCs via bidirectional secure communications. After receiving the commands issued by the DSO, each MGC strategically adjusts LC settings domains to fulfill the DSO's requirements. Besides, MGCs continuously report the operating conditions of individual microgrids (e.g., bidding information, resource availability, operation margins) to help DSOs make credible decisions for enhancing the resilience, reliability, economics,

sustainability, and security of power supplies. Accordingly, the DSO eases the management of the power distribution system with numerous DERs.

Individual microgrids can also vary in design by considering the characteristics of the local power supply and demand. Although alternating current (ac) microgrids are prevalent in contemporary ac-dominant power systems, direct current (dc) microgrids are becoming increasingly more attractive as photovoltaic generation and dc loads (e.g., computers, phones, appliances, LED lighting) proliferate in power systems [50]. DC microgrids not only accommodate dc-type DERs and dc loads directly without incurring additional losses in power conversions, but also facilitate the development of simpler operation and control strategies without any concerns of frequencies and reactive power supplies. Furthermore, technical benefits of ac and dc microgrids can come together in a hybrid ac/dc microgrid, where the ac and dc sections are interfaced by inter-linking converters (ICs) [51].

In terms of distributed power systems, we consider nanogrids as a technologically simpler and smaller version of a microgrid [52], [53], which is embedded within a microgrid for serving critical loads. Provisional microgrids [54] are helpful for boosting the utilization of local DERs together with existing microgrids. Unlike microgrids, provisional microgrids do not possess the islanding capability. Instead, they count on electrically connected microgrids for islanding from the main grid. Hence, expanding the role of microgrids will catalyze the development of distributed power systems that further enhance the operational flexibility and efficiency of the DSO. In general, it is envisioned that power systems will inevitably undergo structural and behavioral changes due to the prevalence of microgrids.

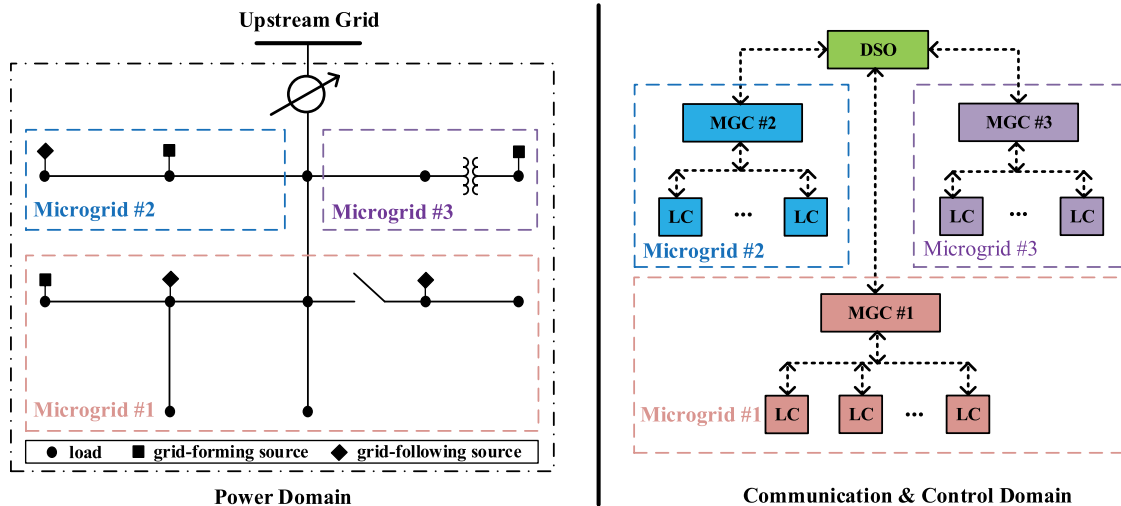


Fig. 6. Microgrids in the IEEE 13-bus feeder.

IV. ENHANCING THE POWER SYSTEM RESILIENCE

A. Microgrid Operations Against Extreme Events

Microgrids are normally operated in parallel with the main grid until the main grid fails to function correctly during an extreme event. The islanding process of a microgrid can be triggered either proactively by the DSO which has perceived the occurrence of a severe disruption in the main grid, or automatically by its MGC which has detected the excursions of frequency or voltage (e.g., voltage exceeding $\pm 5\%$ of nominal value [49]). In such occasions, DERs interfaced by power electronic devices respond quickly to restore the normal microgrid operations without enduring significant transients during islanding [55]. Particularly, microgrid frequency and voltages can be regulated by the grid-forming DERs. In the island mode, the local power supply relies on available DERs, especially those utilizing renewable energy (e.g., wind, sunlight, biomass), in case the flow of natural gas or oil to the microgrid is disrupted. Besides, volatile renewable energy generation can be balanced by utilizing local energy storage devices, as well as the microgrid demand response guided by the MGC. As the last resort, noncritical microgrid loads can be temporarily curtailed for supplying the critical loads without any interruptions.

In this respect, with sufficient preparedness, microgrids can actively disconnect themselves from the main grid, rapidly transfer their operation into the island mode, and deliberately function as an autonomous entity to sustain the local power supply. After isolation from the disrupted main grid, microgrids ride through the extreme event by optimally managing local DERs. Additionally, islanding the microgrids strategically can prevent the deterioration of main grid operating conditions. The islanded microgrids will be reconnected to the main grid seamlessly once the extreme event at the grid side is completely eliminated [58]. The successful resynchronization of an ac microgrid with the main grid necessitates an acceptable range of voltage magnitude, frequency, and phase angle differences at the two ends of PCCs, whereas the reconnection of a dc microgrid to the ac grid through a bidirectional dc/ac conversion will only be subject to voltage magnitude adjustments. The PCC location in a hybrid ac/dc microgrid could have a major impact on islanding and resynchronization processes with the main grid.

Having a sufficient level of generation capacity, the microgrid may act as a virtual feeder for exchanging power with the main grid and picking up critical loads outside its territory. Fig. 7 compares the evolving operating condition in terms of power quality outside and inside microgrids when the associated power distribution system is stirred by an extreme event.

During the extreme event, the system performance suffers a continuous degradation as the disruptions are prolonged. As shown in Fig. 7(a), electricity consumers outside

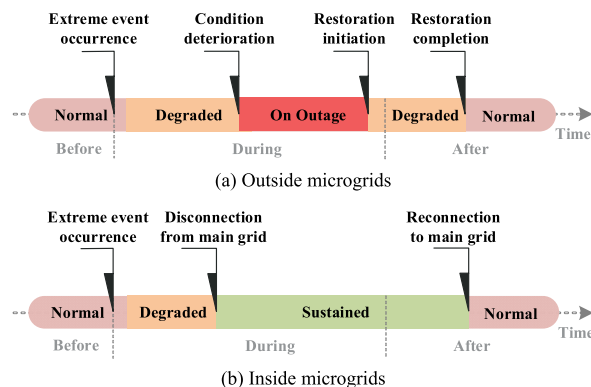


Fig. 7. Evolution of the operating condition.

microgrids will accordingly suffer from power outages and have access to the normal power supply only after the restoration process is completed. On the contrary, those within microgrids will enjoy a more sustained power supply during the extreme event, as shown in Fig. 7(b). In this case, the intentional microgrid islanding provoked by MGCs will protect the local power supply from the deteriorating system condition of the external system. The microgrids can potentially operate in the island mode and provide satisfactory electricity services to local consumers, through the strategic management of on-site DERs, until the main grid is completely restored.

The components within the islanded microgrid are also likely to lose functionality when experiencing an extreme event. As microgrid uses a small control domain, MGC can quickly identify extreme event consequences on microgrid operations and deploy local remedial measures to mitigate microgrid discrepancies in a faster and more effective fashion. Additionally, a loop-based topology provides the microgrid with more opportunities than the traditionally radial configurations to enhance resilience as there will always be an alternative path for delivering power to in-loop loads [56], [57].

Fig. 8 shows the layout of a campus microgrid located at Illinois Institute of Technology (IIT), where the power distribution circuits are divided into seven loops. Each loop includes several DERs and load entities. Assuming that a permanent fault occurs somewhere in a loop, the associated switches will quickly reconfigure the loop connections so that power flows can be automatically rerouted via unaffected paths without interrupting the power supply to loads. However, loads severed through a radial configuration would inevitably suffer an outage when any upstream component fails.

B. Management of Networked Microgrids

The islanded microgrids that are geographically close to each other can be networked to form a microgrid cluster which will function as a single aggregated island

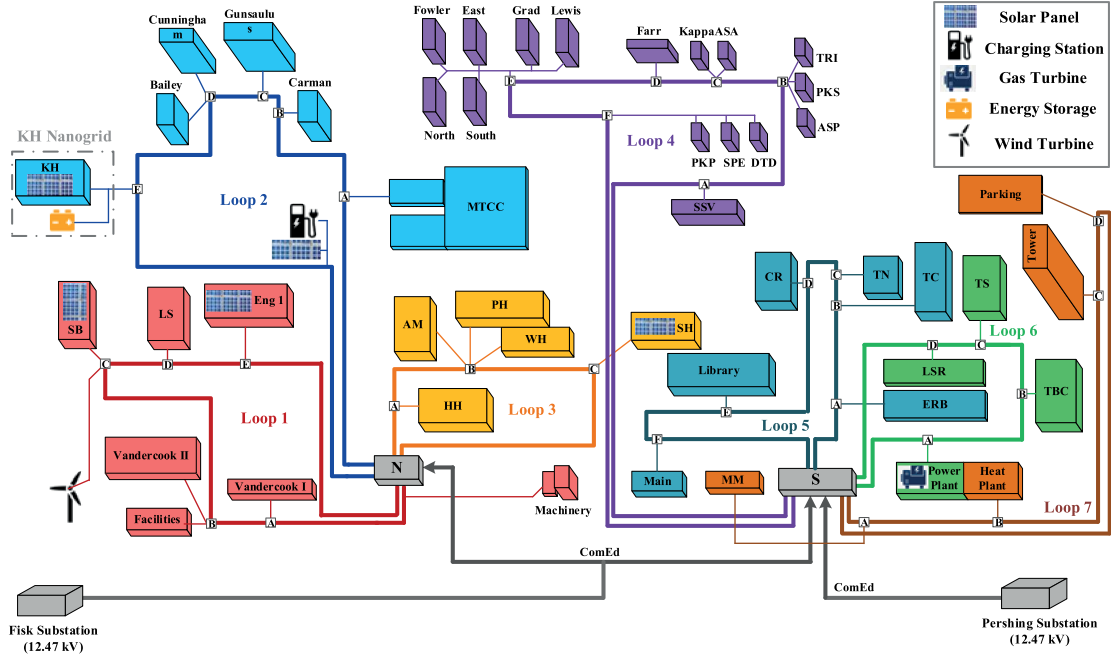


Fig. 8. IIT campus microgrid with the loop-based topology.

from the main grid's viewpoint [59]. Networked microgrids provide a microgrid cluster with a means of gaining additional flexibility for resilient operations. Through steady communications with the DSO, each microgrid can decide when and how to be networked with participating microgrids. Meanwhile, networked microgrids operate cooperatively by sharing critical DER resources in order to overcome potential power deficiencies in microgrid clusters.

Furthermore, each microgrid can have a heterogeneous topological and operational design that can meet the unique requirements of local power supplies. Hence, the potential benefits of microgrids can be realized more sufficiently when they are networked to achieve higher efficiency and security for power supplies. The power transfers among networked microgrids are regulated by the DSO which periodically collects and integrates the information provided by networked microgrids, guides their participation and interconnection in the cluster, and determines the optimal transfer of power for coordinating their operations in a cohesive manner.

Fig. 9 shows a set of networked microgrids where microgrids #1, #2, and #3 are hybrid ac/dc, ac, and dc microgrids, respectively. These microgrids are interlinked by multiple types of ICs which enable power exchanges at the nominal frequency of the main grid. Microgrid #1 is connected through its ac section and thus IC #1 is in essence a transformer. Microgrid #2 is operated at a different frequency, and thus IC #2 is a bidirectional ac/dc/ac converter. Microgrid #3 is linked through IC #3 which is a bidirectional ac/dc converter. Additional interconnection circuits in Fig. 9 which are dedicated to networking the microgrids

are built in parallel with the existing distribution feeder, and the two circuits are interconnected via the coupling point. The redundant design provides the networked microgrids with a higher level of resilience and configuration flexibility against extreme events.

Networked microgrids can also shorten any power supply interruptions outside their territories and expedite restoration processes in distribution power systems. Particularly, microgrids enable power systems to adopt a bottom-up restoration strategy in parallel with the conventional top-down strategies. That is, networked microgrids start energizing the local distribution network by restoring the supply to critical loads within their territories, while the bulk generators start the reenergization of transmission network components. These two restoration strategies are implemented in close coordination until the entire power system is synchronized. Accordingly, the system performance can

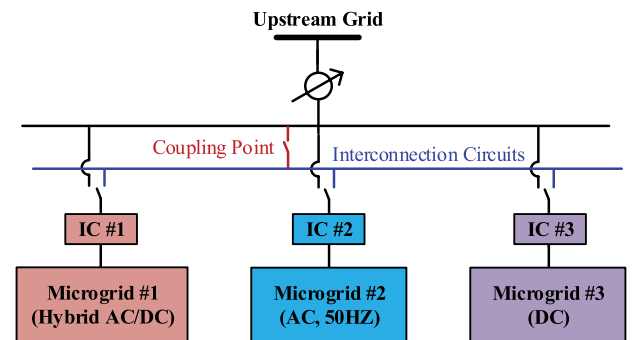


Fig. 9. Configuration of networked microgrids.

be recovered much more rapidly using a simultaneous bidirectional fashion.

The deployment of networked microgrids can establish resilience in power systems to the extent that it would help defy pertinent extreme events in corresponding microgrid regions. The uneven proliferation of damages in extreme events further substantiates the emergence of networked microgrids since the probability that all functional components in the entire microgrid cluster fail to operate simultaneously is rather minute [60], [61].

C. Microgrid-Aided Resilience Enhancement

Since the resilience to an extreme event depends on the inherent characteristics of a power system infrastructure, as well as its operational effectiveness, resilience can be improved by introducing infrastructural reinforcements and operational enhancements. Generally, infrastructural reinforcement comprises hardening and planning measures, whereas operational enhancement can be further divided into preventive and corrective actions, as shown in Fig. 10. Given the unique benefits of distributed power systems in support of the power system resilience, the effectiveness of these measures can be improved with the aid of microgrids.

Infrastructural measures render the electricity infrastructure more resistant to disruptions and easier to be recovered if disrupted. On one hand, hardening aims at strengthening power system components to withstand severe disruptions so that hardened components are likely to retain adequate functionality during extreme events. Relevant measures could include utilizing more underground cables, strengthening power conversion devices, building additional power lines along more secure geographical paths, and fortifying PCCs. On the other hand, planning considers both adding redundancy to the electricity infrastructure and driving the infrastructure to undergo structural changes. In this way, the infrastructure inherently gains more resilience to extreme events. Relevant measures include deploying additional networked microgrids, refining and strategically reconfiguring networked microgrid

topologies, and increasing the redundant generation supply and storage capacity in networked microgrids.

Operational measures are helpful in mitigating the inevitable consequences of power disruptions and realizing the need for the rapid restoration of disrupted supplies. For one thing, preventive measures prepare power systems adequately for potential disruptions before extreme events occur. They can not only lower the influence of disruptions on power system operations, but also reserve an adequate capability for responding to unfolding disruptions. For instance, microgrids can consider potential islanding options as part of their preventive operation strategy so as to achieve smooth transitions into the island mode and subsequently sustain the local power supply during extreme events. Such measures include allocating backup generation and charging up the energy storage located inside networked microgrids, regulating power exchanges with the main grid, and planning intentional microgrid islanding purposefully so that the maximum level of microgrid load can be preserved. For another, corrective measures play a role in responding to and recovering from disruptions. Particularly, such measures will correct the system performance promptly and restore the normal operation effectively so that power systems can ride through extreme events. Such corrective measures include proactively islanding microgrids, networking microgrids to form an effective microgrid cluster, starting up the backup generation, curtailing noncritical loads swiftly, repairing damaged components through self-healing methodologies, and restoring the power supply to disrupted areas by utilizing a practical restoration strategy.

In practice, preventive and corrective actions are complementary. Preventive actions themselves cannot maintain a satisfactory system performance after extreme events if they are unfolded without the help of corrective actions. Meanwhile the effectiveness of corrective actions in response to disruptions will be limited if no preventive action is prescribed and implemented. The synergy of resilience enhancement measures is illustrated in Fig. 11, which shows the transitions among operating points and the variations in resilience regions after the deployment of these measures in a power system.

Given a potentially extreme event, a resilience region is represented by a set of operating points that can offer a desired level of support for attaining resilience. Accordingly, the system achieves sufficient resilience against the extreme event by staying within this region. In Fig. 11, the system is initially operated optimally at point A. With the deployment of infrastructural measures, the optimal operating point traverses to point B due to the fact that infrastructural changes (e.g., introducing additional renewable-based generation capacity in microgrids) can also improve the operational efficiency. Point C represents the operating point with sufficient preparedness against the extreme event after the preventive actions (e.g., minimizing power exchanges with the main grid) are considered. Upon the occurrence



Fig. 10. Measures for resilience enhancement.

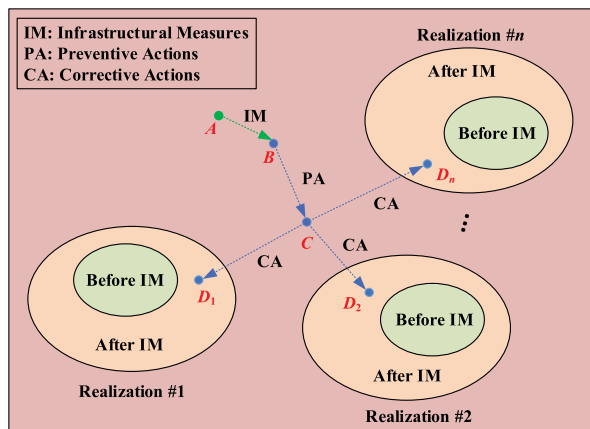


Fig. 11. Synergy of resilience enhancement measures.

of an extreme event, corrective actions (e.g., islanding) are considered as a response to the unfolding disruptions and accordingly the operating point is transferred to the resilience region which has already been expanded by infrastructural measures. Since the virtual transfer distance is rather limited (e.g., by real-time system controllability), point C should be well positioned so that the operating points can be maintained within the resilience region under possible extreme event. However, the position of point C , in turn, depends on the deployment of preventive actions, as well as infrastructural measures. Clearly, a proper and close coordination of these measures can play a significant role in resilience enhancement, while the effectiveness of the deployed measures can be quantified based on the results of resilience analyses discussed in Section II-E.

D. Integrated Framework for Resilience Enhancement

Resilience enhancement in power systems acts as a defense line for guarding the system against various types of natural and man-made threats as planning efforts provide human operators with flexibility in preparing for and responding to extreme events. Given a potentially extreme event, enhancement measures are to be coordinated for boosting the resilience level. In principle, infrastructural measures can facilitate operational measures. Hardening efforts help reduce the needs for introducing costly operational measures since the infrastructure would more likely remain intact following extreme events. On the contrary, operational measures tend to become insignificant in the wake of extreme events if a strong support by a robust infrastructure is lacking; in such case, the system can collapse quickly before human operators can respond to the disruptions effectively.

Meanwhile, infrastructural measures without considering their implications on operational measures may hamper any improvements in resilience. For instance, utilizing underground cables in place of overhead lines makes the

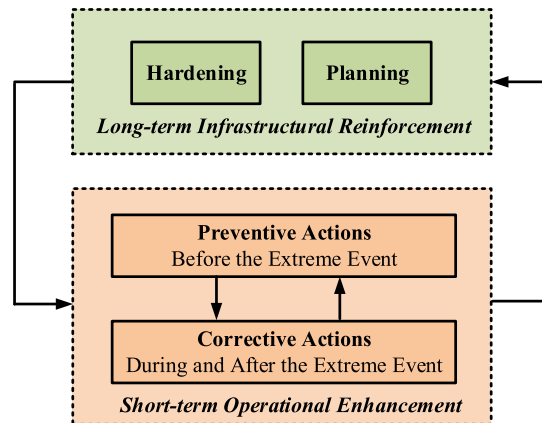


Fig. 12. Integrated decision framework for resilience enhancement.

power supply potentially less susceptible to extreme events but the costly venture may also require substantial efforts for the maintenance of damaged cables in extreme events. Accordingly, resilience may be weakened by the prolonged restoration processes which could be due to the inefficiency of maintenance crew and the lack of sufficient or specialized equipment in extreme conditions.

Fig. 12 depicts an integrated decision framework for determining the most cost-effective solution to achieving the power system resilience in an extreme event. This multi-layer framework takes into account the interdependency of infrastructural and operational measures as well as the interactions of preventive and corrective actions. Additionally, this framework presents an inherent leader-follower relationship between infrastructural reinforcement and operational enforcement, where the former acts first as the leader, followed by the latter with mutual influences on the decision-making process. In order to discover the optimal combination and the sequences for implementing the resilience enhancement measures, stochastic [62] or robust [63] optimization problems can be formulated within this framework after modeling the uncertainty embedded in extreme events and its implications on power system operations.

V. RIDING THROUGH EXTREME EVENTS

A. Droop Characteristics for Load Sharing

Microgrids have the capability to island themselves and function autonomously to maintain the local power supply in the wake of extreme events. This significant feature protects the microgrids from any major disruptions in the main grid, but poses technical challenges on the microgrid operation and control. Practically, each islanded microgrid maintains its frequency and voltages with the help of on-site DERs. These DERs are interfaced predominantly by power electronic devices under the control of LCs. Accordingly, wind turbines and photovoltaic panels are commonly controlled by the maximum power point tracking technique

in all conditions in order to take full advantage of renewable energy [64], which means they act as current sources with only grid-following capabilities. Conversely, DERs with grid-forming capabilities contribute to the regulation of frequency and voltages, when they are controlled as voltage sources with droop characteristics. A droop characteristic, which resembles the frequency regulation posed by the governor of a synchronous generator, is the locus of steady-state operation points. Droop characteristics help realize a fast and decentralized load sharing under various conditions. Additionally, droop characteristics make the desired load sharing among grid-forming DERs (referred to as DERs hereinafter) dependent merely on local measurements.

Fig. 13 illustrates the role of droop characteristics in an ac microgrid hosting two DERs. Here, the application of droop characteristics to regulate frequency and voltages is ubiquitous [65]. Conceptually, any increases in real power demand is proportionally allocated among DERs at a lower system frequency, while the share of the increased demand for reactive power among DERs potentially decreases local voltage magnitudes. In other words, if a sudden power imbalance occurs, the microgrid will experience some changes in its frequency and voltages, while the participating DERs help regulate the frequency and local voltages according to their droop characteristics. Mathematically, these droop characteristics are expressed as

$$f = f_n - k_p(P_{ac} - P_{ac,n})$$

$$V_{ac} = V_{ac,n} - k_Q(Q_{ac} - Q_{ac,n})$$

where f and V_{ac} are the frequency and terminal voltage magnitude measured by the LC; P_{ac} and Q_{ac} are the measured real and reactive power output by the DER; f_n , $V_{ac,n}$, $P_{ac,n}$, and $Q_{ac,n}$ are the nominal values of f , V_{ac} , P_{ac} , and Q_{ac} , respectively; and k_p and k_Q are the positive droop coefficients.

In Fig. 13, the coherent response of the two DERs can be equivalently modeled as an aggregated droop curve at the microgrid level after the superposition of individual droop characteristics at the DER level. Initially, DERs #1 and #2 steadily generate power $P_{1,n}$ and $P_{2,n}$ at the nominal

frequency f_n , respectively. A sudden increase in real power consumption results in an immediate decrease in the microgrid frequency, followed by a generation increase in both DERs. The power imbalance is eventually offset by changes in DER power generation (reflected by the aggregated droop characteristics) as well as loads (due to its frequency-sensitive characteristics). At the new steady state, DERs #1 and #2 generate power P'_1 and P'_2 at a lower frequency f' , respectively. Clearly, the droop characteristics drive two DERs to share the change in load proportionally.

In dc microgrids, a similar droop characteristic [66] is utilized, which leads DERs to increase their power outputs when the power imbalance results in decreased local voltage magnitudes

$$V_{dc} = V_{dc,n} - m_p(P_{dc} - P_{dc,n})$$

where V_{dc} and P_{dc} are the terminal voltage magnitude and DER power output measured by the LC; $V_{dc,n}$ and $P_{dc,n}$ are the nominal values of V_{dc} and P_{dc} ; and m_p is the positive droop coefficient.

B. Conceptual Load Sharing in Hybrid AC/DC Microgrids

Load sharing, however, is more complicated in a hybrid ac/dc microgrid which needs to maintain the frequency as well as bus voltages in the ac and dc sections. The load sharing in the ac section or the dc section of a hybrid ac/dc microgrid resembles that in an ac microgrid or a dc microgrid. We apply droop characteristics to enable automatic load sharing between ac and dc sections so that the potential benefits of the complex hybrid ac/dc structure can be realized.

In principle, the operations of ac and dc sections are tightly coupled by bidirectional power flows, which are regulated by IC. When a section suffers a power imbalance, the IC enables the other section to assist in restoring the power balance by means of a power transfer. Additionally, each section only needs to interact with its IC for requesting additional power support. The effect of the other section

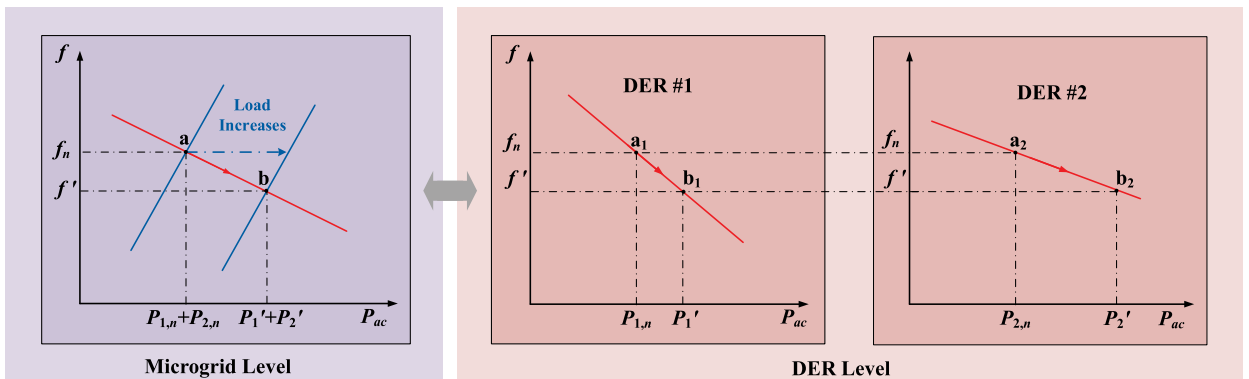


Fig. 13. Droop-based load sharing mechanism.

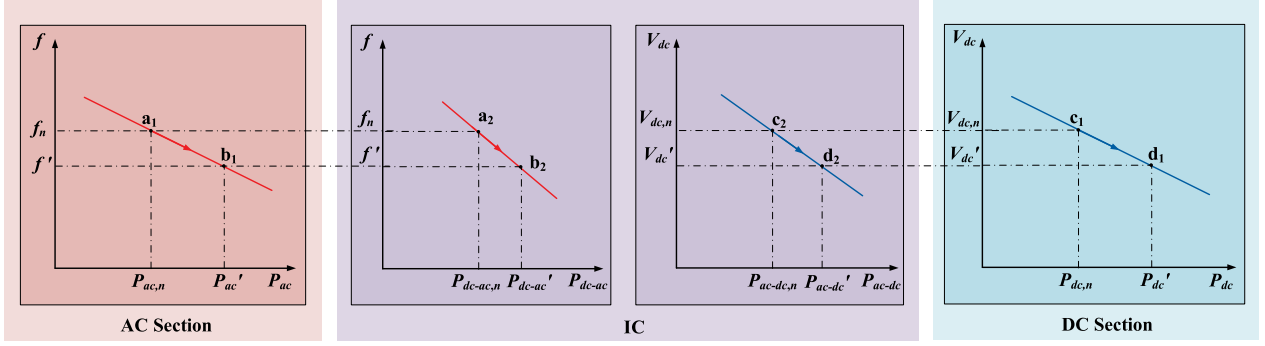


Fig. 14. Partial droop characteristics of IC.

on operation can therefore be simply represented by power exchanges (i.e., power transferred from/to the other section) through IC. In this sense, IC is virtually divided into two parts where each part negotiates independently with its corresponding section for regulating the power transfer. Hence, IC regulates power transfers in both directions by adhering to the following partial droop characteristics [67]:

$$\begin{aligned} P_{dc-ac} &= P_{dc-ac,n} - n_{dc-dc}(f - f_n) \\ P_{ac-dc} &= P_{ac-dc,n} - n_{ac-dc}(V_{dc} - V_{dc,n}) \end{aligned}$$

where P_{dc-ac} and P_{ac-dc} stand for the amount of power transferred from dc to ac section, and from ac to dc section, respectively; f and V_{dc} are the frequency and voltage magnitude measured at ac and dc terminals of IC, respectively; $P_{dc-ac,n}$, $P_{ac-dc,n}$, f_n , and $V_{dc,n}$ are nominal values of P_{dc-ac} , P_{ac-dc} , f , and V_{dc} , respectively; and n_{dc-dc} and n_{ac-dc} are two positive droop coefficients.

These partial droop characteristics enable IC to respond to any degradations in either section's operating condition based on the terminal measurements. The frequency measured at the ac terminal directly reflects the operating condition in the ac section, while the voltage magnitude measured at the dc terminal is a surrogate measure of power surplus/shortage in the dc section. Particularly, the ac section features an equivalent frequency droop curve that integrates all DERs' frequency regulation characteristics and all loads' frequency sensitivity characteristics in that section, whereas the dc section possesses an equivalent voltage droop curve after integrating the voltage regulation or sensitivity characteristics of DERs or loads that are highly correlated with the dc-terminal voltage of the IC.

Fig. 14 illustrates the load sharing mechanism enabled by these partial droop characteristics. The ac section with a real power shortage will request the additional power injection from IC and meanwhile raises its on-site DER generation according to the individual droop characteristics. Eventually, the frequency will be stabilized at a lower value determined jointly by the equivalent frequency droop curve and the partial droop characteristic regulating the power transferred into the ac section. Similarly, when the

dc section suffers a power shortage, the decreased voltage forces the related DERs and the IC to provide more generation. Eventually, the voltage magnitude measured at the dc terminal of IC will be maintained at a reduced value determined jointly by the equivalent voltage droop curve and the partial droop characteristic regulating the power transferred into the dc section. In such cases, IC interacts with individual sections separately without considering the interdependency of their respective operating conditions.

C. Synthetic Droop Characteristics

Since power transfers cannot occur in both directions simultaneously, IC needs to determine the flow direction before realizing the power transfer. In essence, the two partial droop characteristics could have a conflict due to the distinct direction of regulated power transfer. Each partial droop is biased toward a power transfer that can improve the associated section's operating condition. However, IC must coordinate partial droop characteristics for the power transfers requested by both sections and assign different weights to the droops for coordination. The resulting power transfer is expressed as

$$P_{ic} = \alpha P_{dc-ac} - \beta P_{ac-dc}$$

where P_{ic} is the desired power transferred via IC, and the reference direction is assumed from the dc section to the ac section; α and β are two nonnegative constants representing the weights; if a section's weight is equal to zero, the power transfer is independent of the corresponding operating condition; if a section has a much larger weight, the power transfer is dominated by the corresponding operating condition.

Accordingly, the two partial droop characteristics in a hybrid ac/dc microgrid are integrated as a synthetic droop characteristic for coordinating the operations of the two sections, using the following substitutions:

$$\begin{aligned} P_{ic,n} &= \alpha P_{dc-ac,n} - \beta P_{ac-dc,n}, \quad n_{ic} = \frac{1}{\alpha n_{dc-ac}} \\ E &= f - \frac{\beta n_{ac-dc}}{\alpha n_{dc-ac}} V_{dc}, \quad E_n = f_n - \frac{\beta n_{ac-dc}}{\alpha n_{dc-ac}} V_{dc,n} \end{aligned}$$

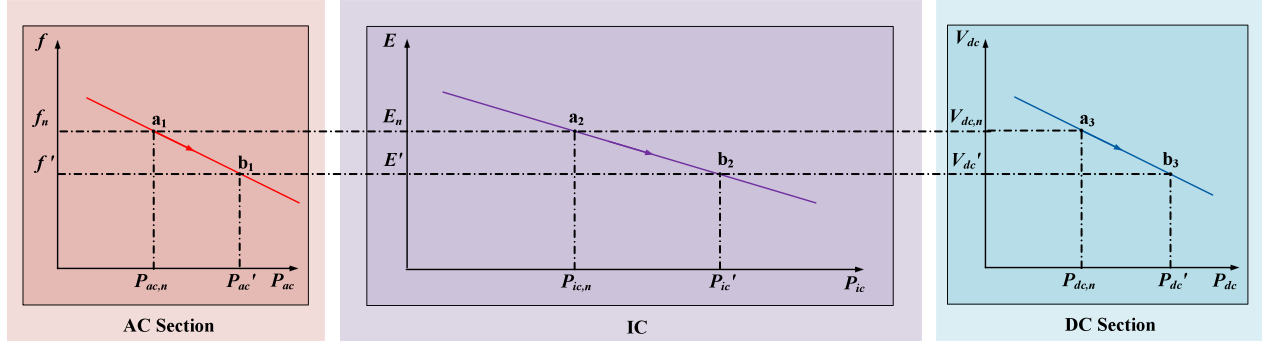


Fig. 15. Synthetic droop characteristic of IC.

where E denotes the performance difference between the two sections in terms of the IC's terminal measurements f and V_{dc} ; $P_{ic,n}$ and E_n are the nominal values of P_{ic} and E ; and n_{ic} is the equivalent droop coefficient. Finally, the synthetic droop characteristic governing the IC flow is expressed as

$$E = E_n - n_{ic}(P_{ic} - P_{ic,n}).$$

Fig. 15 illustrates the synthetic droop characteristic for regulating the intersection power transfer when the ac section has a real power shortage. In this case, the frequency deviation in the ac section immediately results in a larger transfer of real power from the dc section to the ac section. To meet the larger power extraction, the dc section will increase its power generation which results in a lower dc-terminal voltage of the IC. Once the power transfer is stabilized, the operating condition of the hybrid ac/dc microgrid reaches a new equilibrium where the ac power is shared proportionally between ac and dc sections. However, without the additional power support from the dc section, the ac section may suffer a significant frequency drop or severe frequency instability.

D. Hierarchical Control Architecture

Microgrids may face various uncertainties during extreme events which include random component outages and hourly forecasting errors for loads, renewable energy, and fuel availability for power generation, resulting in additional complications in microgrid operations. Once an uncertainty scenario is realized, the droop-based load sharing mechanism provides microgrids with the ability to react promptly to power imbalances. Such mechanism offers DERs the plug-and-play capability that introduces additional operational flexibility for microgrids to respond to and recover from power disruptions. However, droop-based load sharing will result in inevitable regulation errors represented by signal deviations from nominal values. Thus, additional control strategies are considered to safeguard microgrids against challenges posed by extreme events.

A trilevel hierarchical control architecture, shown in Fig. 16, presents a viable resilience solution for any type of

microgrids including ac [68]–[70], dc [71]–[75], and hybrid ac/dc [76], [77]. Table 2 compares the three levels of the hierarchical control architecture. Primary control realizes fast load sharing among DERs, which is performed by LCs in a decentralized manner. Secondary control corrects regulation errors of primary control, which is performed by LCs (possibly under the MGC direction). Tertiary control aims at achieving the optimal energy management which is a centralized function in the MGC.

Primary control relies on droop characteristics for automatic load sharing. Mathematically, droop characteristics are generalized as follows:

$$y = y_n - \lambda \cdot (x - x_n)$$

where x and y are two correlated variables, and y can be regulated by determining x for a control entity; x_n and y_n are the nominal values of x and y , respectively; and λ is the droop coefficient with a positive value. For different control entities, x , y , and λ represent different meanings for determining the droop characteristics. Table 3 summarizes the typical droop characteristics in networked microgrids. In the following, the hierarchical control is analyzed in terms of the generalized droop characteristic.

Secondary control plays a significant part after primary control has stabilized the microgrid frequency. Secondary

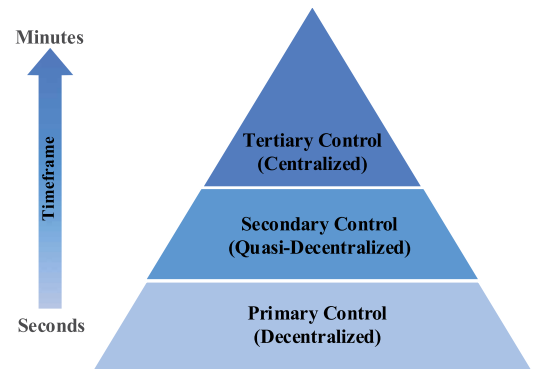


Fig. 16. Hierarchical control architecture.

Table 2 Comparisons of Control Levels

Level	Responsibility	Implementation
Primary control	Fast load sharing	LCs
Secondary control	Regular error correction	LCs + MGC
Tertiary control	Optimal energy management	MGC

control aims at mitigating the regulation errors introduced by primary control. When a microgrid is connected to the microgrid cluster, the power exchange with the cluster should match the scheduled level as required by the DSO in order to guarantee that any power imbalance in the microgrid is locally mitigated without influencing the operation of other networked microgrids. In this sense, microgrid control error (MCE), which is defined as the sum of mismatched power transferred to the cluster, will serve to indicate whether the microgrid generation must be raised or lowered. If the MCE is positive/negative, then the microgrid must raise/lower its total generation. The MCE is collectively offset by the selected DERs in a predefined manner (e.g., according to the principle of proximity). The MGC at each networked microgrid monitors the MCE in real time and notifies LCs when the MCE suffers a sudden change. In case the microgrid is incapable of satisfying the scheduled power exchanges, the MGC will notify the DSO for rescheduling the power exchanges pertaining to the microgrid constraints.

Accordingly, there may exist two types of regulation errors once primary control has stabilized the microgrid: the deviation of x determined by the droop characteristics, and the MCE caused by the power exchange mismatch resulting from the change in x after primary control. The former is locally measured by LCs, while the latter is signaled by the MGC. If a DER is not participating in offsetting the MCE, secondary control dependent only on the LCs can be performed in a decentralized manner. When secondary control is completed, the droop characteristic for that DER will be modified as

$$y = y_n - k \cdot (x - x')$$

where x' is the value of x stabilized by primary control before secondary control is started. Otherwise, secondary control performed the LCs as signaled by MGC. When secondary control is completed, the droop characteristic for that DER will be modified as

$$y = y_n - k \cdot (x - x^\circ)$$

Table 3 Typical Droop Characteristics

Control Entity	x	y	λ
DERs in AC microgrids/sections	P_{ac}	f	k_P
	Q	V_{ac}	k_Q
DERs in DC microgrids/sections	P_{dc}	V_{dc}	m_P
IC	P_{ic}	E	n_{ic}

where x° is the new reference value of x , expressed as

$$x^\circ = x' + \rho \text{MCE}$$

and MCE represents the MCE signaled by the MGC; ρ is the participation factor indicating the percentage that DER takes in offsetting the MCE.

Tertiary control is performed routinely by the MGC or in an event-driven fashion with the purpose of achieving the economic and secure operation in the microgrid. Through communications with the DSO, the MGC can decide when to disconnect or reconnect with the main grid, as well as when to join or quit the microgrid cluster. The implications of the external power system on microgrid operations can therefore be perceived before the MGC initiates the optimal microgrid-wide energy management. Specifically, DER settings and power exchanges with external power systems can be optimized by performing security-constrained economic dispatch under various conditions in a look-ahead manner [78]–[80]. When tertiary control is completed, the DER droop characteristic will be updated as

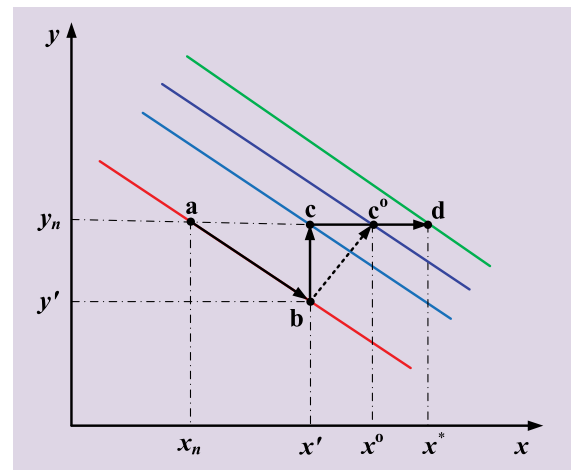
$$y = y_n - k \cdot (x - x^*)$$

where x^* is the desired value of x for optimal energy management.

In practice, both secondary and tertiary controls are accomplished by applying proportional, integral, and derivative controllers for error compensation. Additionally, the effectiveness of hierarchical control of IC depends on the successful execution of hierarchical control in both ac and dc sections in a hybrid ac/dc microgrid.

E. Illustration of Hierarchical Control

Fig. 17 shows hierarchical control applied to an entity. The entity is initially operated at point a , corresponding to the nominal x_n and y_n values. Once y suffers a sudden


Fig. 17. General process of hierarchical control.

decrease, x will be forced by primary control to be adjusted. Upon the completion of primary control, the operating point will move to point b on the droop curve, when x is raised to a higher value x' at a lower value y' . Secondary control will then start to restore y to its nominal value y_n . If this control entity participates in mitigating the MCE, the operating point will be transferred to c° corresponding to (x°, y_n) . Otherwise, the operating point will be c , where x remains its value x' after y achieves the value y_n . The operating point will be retained until the MGC assigns an updated value x^* at the same value of y . After the execution of tertiary control, the operating point will eventually move from c° (or c) to d as its final abode. Obviously, primary control causes the steady-state operating point to migrate along the droop curve, whereas secondary and tertiary controls result in the shift of the droop curve. This hierarchical control architecture enhances, through its distinguished error mitigation and online adjustment capabilities, the performance of the fine-grained microgrid operation and control. Particularly, hierarchical control facilitates a smooth islanding from and resynchronization to the main grid or other microgrids in the cluster.

Suppose the islanding causes a frequency drop in an ac microgrid due to a mismatch between the local load and generation. Accordingly, LCs perform primary control to limit the frequency drop by raising the DERs' power outputs. After the completion of the load sharing in primary control, LCs initiate secondary control for restoring the frequency to its nominal value. However, if the frequency continues to deteriorate, MGC will trigger the underfrequency load shedding so as to mitigate the threat of frequency collapse. The curtailed loads will be partially or fully restored through tertiary control once the microgrid has the sufficient capability to supply power to them without an adverse effect on its operation. Accordingly, hierarchical control allows a microgrid to maintain its frequency within the allowable range and contribute to a successful transition to island mode.

VI. RECOGNIZING THE MERITS OF ICT

A. Improved Situational Awareness

Situational awareness is a critical factor in strengthening the power system resilience to various extreme events [81]. It provides effective guidance on how to protect power systems adequately from disruptions before the realization of such events, and how to reduce the implications significantly during and after their occurrences. Conversely, insufficient situational awareness could result in deficient preparation for, and incorrect response to extreme events. Such inadequate actions, which are often identified as one of the major causes of massive power outages across the globe, can position power systems in a serious danger [82].

Sufficient situational awareness in microgrids requires the full observability of current situations and the intense visibility into changing operating conditions. Basically,

human operators should always be aware of when and where an extreme event is evolving (e.g., utilizing methods like state estimation) and how it may disrupt microgrid operations (e.g., utilizing methods like real-time contingency analyses). Meanwhile, human operators should exploit the look-ahead capability to implement effective measures in response to disruptions in any extreme events.

ICTs, especially metering and monitoring techniques, are advancing rapidly for power system applications, which open the door to drastically increase the situational awareness in microgrids. The deployment of a monitoring scheme with full coverage necessitates the installation of a host of sensors with acceptable precision and accuracy, which encompass dispersed components located throughout the microgrid, including field devices, feeders, and load entities [83]. To maintain observability during extreme events, redundant sensors are to be deployed as backup. Using real-time measurements, human operators will be able to better observe changes evolved in operating conditions and take actions with more confidence which would result in suitable operation criteria. For instance, sensors can simplify the detection and the identification of malfunctions or failures in field devices during extreme events, by sensing ambient data (e.g., temperature, humidity) and operating states (e.g., voltage, current).

The automatic outage reporting capability of smart meters [84] makes it particularly convenient to pinpoint faults and identify the outage scope so that operators can direct more effective efforts to restore electricity services in microgrids. Smart meters also assist operators in guiding consumers for emergency demand response pertaining to extreme events. Besides, the advent of phasor measurement units (PMUs) [85] meets the requirements for observing fast system dynamics initiated by system disruptions. In every second, PMUs typically provide 10–60 synchronized high-resolution measurements, which could equip operators with sufficient information on evolving operating conditions during extreme events [86]. PMU data are also applicable to improving the system efficiency and credibility in postcontingency resilience analyses [87].

Clearly, situational awareness in microgrids is the catalyst for making effective predictions and timely decisions concerning extreme events. In such circumstances, it is also important to stress the necessity of an efficient data management strategy for monitoring, validating, and interpreting the collected information from a variety of sources. Typical applications include big data management [88], real-time event detection [89], and user-friendly visualization [90]. Only when the applicable data are integrated and processed efficiently can operators gain the capability to fully understand and effectively mitigate the imminent threats following extreme events. The improved situational awareness also gives birth to finer grained resilience-oriented applications for microgrid operations, like the early warning system for projecting any system deteriorations.

B. Self-Adaptive Communication

The utilization of sophisticated ICTs facilitates a rapid and reliable communication network that is essential for the management of microgrids, particularly in response to extreme events. Most of the existing communication networks for monitoring, control, and management of microgrids are realized with a predesigned network functionality (e.g., traffic routing) [91]. Accordingly, it is rather cumbersome and even impractical to reconfigure the existing functionality against any sudden changes in a communication network (e.g., link failure, link congestion). In practice, employing such a nonadaptive paradigm may have an inevitably adverse impact on microgrid operations (e.g., delayed response and reduced situational awareness due to link failures) during extreme events, and may even become a bottleneck for improving the power system resilience [92].

The communication diagram based on software-defined networking (SDN) is a viable option for mitigating excessive limitations and providing distinguished resilience supports to communication architectures. Technically, SDN is an emerging design that makes the control of a communication network directly programmable by transferring the network's control logic from underlying switches to a logically centralized controller. Accordingly, switches in an SDN-enabled network are simply in charge of forwarding traffic with rules that are dynamically configured in their flow tables by the centralized controller through open application programming interfaces [93].

Benefiting from the separation of control and data planes, the SDN simplifies the network configuration while enforcing security policies. When implemented in microgrids, the SDN facilitates the networking of various components through the runtime programmability, and improves the traffic management to better support microgrid applications by means of global visibility. Meanwhile, the SDN enables the refinement of communication technologies for microgrid operations by keeping the portion that works and eliminating the portion that does not. Particularly, the SDN controller with direct network programmability can communicate and cooperate with the centralized MGC so as to catalyze innovative microgrid applications for stretching out the resilience paradigm. Fig. 18 depicts the hierarchical architecture of an SDN-enabled microgrid.

In general, SDN technologies bring the following benefits to prepare communication networks adequately for playing a vital role in facilitating the resilience in microgrid operations.

- The SDN empowers communication networks to evaluate and change their own behaviors at runtime so as to quickly adapt themselves to evolving networking conditions. When a microgrid is experiencing an extreme event, SDN offers the capability to continuously detect, analyze, and mitigate the

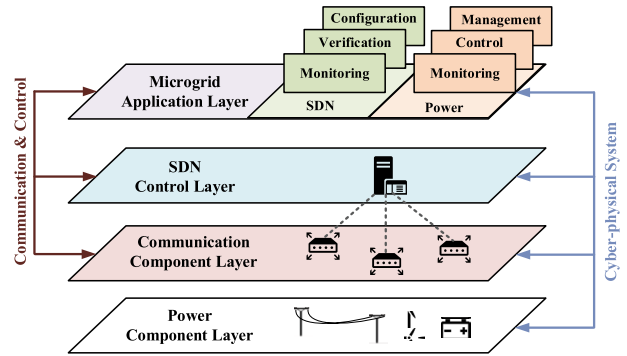


Fig. 18. SDN-enabled cyber-physical microgrid architecture.

implications in the communication network. For instance, the SDN enables dynamic reconfiguration by strategically reestablishing the routing that can avoid disrupted or compromised switches and communication links. Hence, the SDN helps maintain a reliable and rapid data transmission for microgrid applications. In order to realize a fast failover, the SDN controller can install backup paths directly into the switches in case a certain emergency occurs [94].

- The SDN ends the confusion about where to deploy effective defensive measures for preserving the functionality of a communication network, since the SDN controller plays the most important role. If the SDN controller is well protected from potential disruptions, the communication network under control will always have a chance to survive. In order to avoid a single point of failure, an extra SDN controller is deployed as a backup for the primary controller, and control functions are implemented by multiple controllers in a distributed but coordinated fashion.
- The SDN provides a mechanism for building a virtual network layer on the top of the physical communication network in order to meet diverse quality-of-service (QoS) requirements of data transmission for different microgrid applications [95]. In fact, control commands display more stringent requirements for timeliness than measurements from meters. The latency or the loss of control commands may directly lead to insecure microgrid operations. Meanwhile, missing or mislabeled measurements will affect the situational awareness in critical conditions and slow down the decision-making process. In essence, the virtual network layer consists of multiple virtual local-area networks (VLANs) in accordance with diverse QoS requirements. In each VLAN, the SDN controller has access to real-time network traffic monitoring

data for evaluating and optimizing the QoS more effectively. Once any underlying networking component is disrupted, the SDN controller will strategically adjust VLAN domains for maintaining the highest level of QoS performance without hampering designated microgrid applications.

In sum, SDN technologies can significantly enhance the performance of communication networks for microgrid operations in extreme events. By adopting the SDN, microgrid communication networks gain greater flexibility in implementing responsive measures against disruptions while meeting the desired communication requirements in a self-adaptive fashion. We should point out that the SDN potential for ensuring the power system resilience ought to be further explored.

VII. MOVING FORWARD: OUTLOOK FOR A HIGHER RESILIENCE

A. Regulations and Policies

Flexible operation modes of microgrids enable power systems to be less susceptible to extreme events by providing additional opportunities to ride through such events successfully. However, bulk power systems cannot yet realize full potentials offered by networked microgrids for resilience enhancement because specific regulations and policies are missing for allowing microgrids to cooperate in extreme events. Commonly, microgrids are operated in accordance with the prevailing interconnection agreements, i.e., IEEE Standard 1547 [49], [96]–[98], which was initially developed for regulating a low penetration level of DERs in power systems. Given the increasingly higher penetration of DERs hosted by microgrids, the suitability of existing regulations is questioned by stakeholders, especially when microgrids are networked to assume the burden of enhancing the power system resilience.

For safety concerns, existing standards prohibit DERs from energizing the main grid during emergencies. Accordingly, microgrids with redundant generation capacity are not allowed to pick up loads that are located outside their territories in the wake of extreme events. In addition, existing regulations are deficient in governing the economics of energy exchanges between microgrids, which directly impede the formation of a microgrid cluster in a cohesive manner for managing extreme events.

We should issue new regulations and policies that can fulfill the growing needs of leveraging networked microgrids in response to extreme events, especially for aiding the main grid in power restoration. The regulations and policies that facilitate the interconnection of microgrids should also provide guidance on overcoming the resilience-oriented technical challenges in power systems. For instance, the capacity of energy storage should account for a well-defined percentage of the total generation capacity in each microgrid so that

networked microgrids can ensure significant benefits for both local electricity consumers and the main grid in the face of extreme events. Actually, energy storage can play a vital role in accommodating the variability of renewable energy sources and maintaining the satisfactory power supply in emergency conditions. Through these efforts, power systems can eventually boost their resilience by tapping into microgrid potential to defeat the might of extreme events.

B. Microgrids as Energy Hubs

The development of modern societies largely relies on harnessing various forms of energy (e.g., electricity, natural gas, water, hydrogen), among which electricity plays a critical role. Given that synergies among various forms of energy can represent a great opportunity for improving social welfare, microgrids may potentially assume the burden of integrating, coordinating and operating multiple energy carriers [99]. Each microgrid will accordingly assume a greater role in an energy hub, shown in Fig. 19, including a variety of energy carriers (i.e., electricity, heat, natural gas) which can be optimized concurrently for satisfying specific objectives [100]–[102].

Within an energy hub, the conversion, transmission, distribution, and consumption of multiple energy carriers are tightly coupled and considered as interdependent. It is also important to stress that the development of microgrids toward achieving energy hubs is in line with the vision of Smart Cities [103], where the management of multiple interdependent energy carriers is coordinated for achieving a higher efficiency and better reliability in metropolitan areas. In this way, the power system resilience will not depend solely on the electricity infrastructure but is also a function of the resilience in other infrastructures where the energy can be converted to electricity and utilized in an optimal fashion [104].

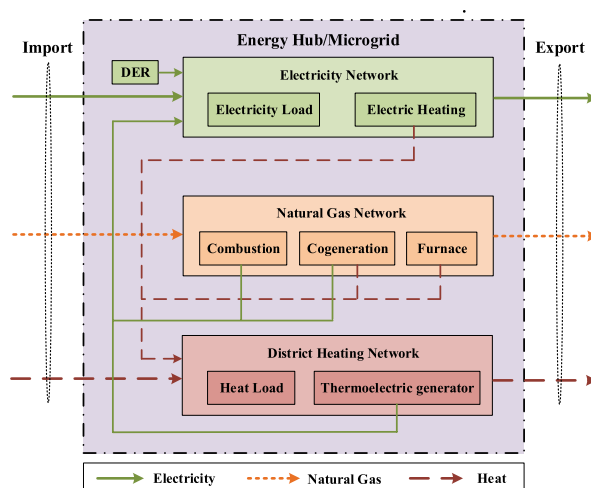


Fig. 19. Microgrid as an energy hub.

Noticeably, strong interdependencies among multiple energy carriers within an energy hub may become a two-edged sword. On one hand, the electricity demand can be optionally supplied by dispatchable units powered by alternative energy, which commonly possesses grid-forming capabilities to assist microgrids in riding through and recovering from extreme events. On the other hand, the impacts of extreme events are likely to spread far beyond power systems, probably resulting in further disruptions and even cascading failures among multiple energy carriers. Consider that an extreme event could damage a natural gas compressor, which ceases the natural gas flow and renders the downstream gas turbine out of options for operation; the resulting power outages could force other gas compressors to stop working. Such component failures may happen recursively between power and natural gas networks, leading to the widespread cross-infrastructure disruptions in an energy hub. After all, a successful migration from microgrids to energy hubs will necessitate multidisciplinary and multipronged efforts for developing sophisticated operation and control strategies that can preserve the resilience of metropolitan systems to various extreme events.

C. Cyber-Physical Security

As microgrids are heavily dependent on ICTs to achieve higher efficiency and better reliability, cybersecurity is playing an increasingly prominent role in complicating the efforts for enhancing the power system resilience [105]. For instance, the deployment of wireless sensor networks for improving the situational awareness in networked microgrids exposes an increased number of access points to cyber intrusions, thereby introducing exacerbated cyber threats. The resulting cyber insecurity would potentially hamper the observability and the controllability of microgrid operations and contribute to the intensified implications of power disruptions in the wake of an extreme event.

Cyber threats affecting networked microgrid operations can be classified into the following three categories.

- 1) Denial of service. When communications between the MGC and LCs (or those between the MGC and the DSO) are blocked or delayed in a microgrid, the microgrid may fail to maintain its normal functionality (or play its role in the microgrid cluster) due to the lack of supervisory information.
- 2) Corrupted measurements. The MGC relies on real-time measurements reported by the field devices for achieving situational awareness and making operational decisions. Accordingly, any integrity violation of measurements may lead to an inaccurate perception of the microgrid operating condition and subsequent improper decisions that jeopardize the efficiency and security of microgrid operations.
- 3) Compromised control commands. The power output settings of DERs are continuously determined and updated by the hierarchical control strategy. When a

control command for a certain DER is tampered, the resulting power output will potentially incur severe instability issues in networked microgrids.

Strong cyber-physical interdependency makes it easy for cyber intrusions to cause serious damages to physical components in microgrids. In particular, cyberattacks can act as a powerful weapon in conjunction with physical contingencies. For instance, cyberattacks during the course of a natural disaster, by jeopardizing the integrity of communication networks, can delay the response to local power outages in networked microgrids. Thus, cybersecurity will be a heightened concern for sustaining the power supply. Efficient cybersecurity management that can meet the unique characteristics of microgrid operations is vital for addressing the evolving cyber threats in the process of building resilient power systems.

VIII. CONCLUSION

Resilience, as a key attribute of Smart Grid, is central to the efforts into modernizing power systems in various parts of the world. Resilience studies will guarantee that the electricity will be available to local residents in case of a widespread outage which could occur due to various extreme events. It is evident that the deployment of microgrids is a promising and all-purpose solution to meeting the public demand for the infrastructure resilience. Generally, the installation of networked microgrids in critical regions will assist power systems in planning for, reacting to, and recovering from extensive disruptions which could be either man-made such as a deliberate attack or nature-made such as a severe weather. Microgrids therefore support the integrity of power systems for enhancing the economic, resilience, security, sustainability, and reliability of the electricity infrastructure and strengthen the power system particularly in critical conditions for defying a variety of extreme events. Although there are still a whole host of regulatory and technical issues that restrict the proliferation of microgrids, the article aimed at substantiating the role of networked microgrids and demonstrating to stakeholders that such installations save lives and enhance the stated mission of public infrastructures. Such installations benefit from the grid supply, local DERs, and clean power generation, and customer participation. It is conceivable that the development of microgrids will drive the conventionally centralized systems to migrate toward distributed and local systems, together with the significant changes in power system management. Eventually, through such installations, Smart Grid will evolve as a solution for promoting microgrid-based distributed systems. ■

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